

HARP: ExactFrac-Constrained Score Serving for GNN Prediction APIs

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Abstract—High-volume GNN prediction APIs serve softmax scores for ranking, routing, A/B tests, and compliance pipelines on graphs from thousands to millions of nodes; production stacks cache by request hash and re-query the same entities under continuous traffic. Operators sometimes also need a positive fraction of raw model posteriors—unmodified $\hat{p}_v$ for calibration and downstream score consumers—which is stricter than replaying a previously sampled noisy response from a cache or ledger. Fresh uniform release noise cannot meet that raw-exact SLA; global caches and seeded noise raise ReplayFrac without restoring ExactFrac $\equiv$ RawExactFrac. HARP is a Big-Data score-serving layer for this constrained setting: constructors rank nodes, hop expansion forms a protected set, a protector runs only there, and validation-time Frac search targets Acc under ExactFrac $\geq c$ (study default $c=0.60$) with an optional audit dial τ . On defense-aware LiRA ($n_{\text{shadows}}=16$), HARP is feasible for $c>0$ with competitive Acc/ECE on large-graph serving probes; residual clean-slice risk is reported as a structural limit of ExactFrac > 0 .

Index Terms—big data prediction APIs, graph neural networks, response caching, ExactFrac, selective score serving, membership audit

I. INTRODUCTION

A. Motivation

GNN prediction APIs return softmax vectors for triage, ranking, and analytics at Big-Data scale [9], [10], [28]: graphs with 10^3 – 10^6 + nodes, high query velocity (clients re-query the same entities), and veracity requirements for downstream score consumers. Production serving stacks hash requests and cache responses to cut latency under repeated queries [40], [41]. Released scores also leak training membership [1], [2], [4], [5]. This paper is about the *systems* intersection of those facts: how to release scores under a raw-posterior service contract while still spending noise where an audit requires it—not a new membership-inference theory paper.

a) Why ExactFrac is operationally necessary.: Two contracts are easy to confuse. *ReplayFrac* asks whether two queries return the same vector (a cache hit or sticky/seeded draw can satisfy this with a *noisy* first answer). *ExactFrac* $\equiv$ *RawExactFrac* asks whether the released vector equals the unmodified model posterior $\hat{p}_v$. Raw equality matters when downstream systems treat API scores as model truth: temperature calibration, threshold routers that must match

offline evaluation, A/B analytics that join on score bits, and compliance pipelines that attest “this is the model’s output.” Caching a noisy draw restores ReplayFrac and QPS; it does *not* restore ExactFrac. We therefore study

$$\max_{\Pi} \text{Acc}(\Pi) \quad \text{s.t.} \quad \text{ExactFrac}(\Pi) \geq c, \quad \text{LiRA}(\Pi) \leq \tau$$

(optional τ), with study default $c=0.60$ as an operator dial (Fig. 2). Fresh uniform LBP has both fractions ≈ 0 (Remark 1). Table VIII shows global-cache and seeded LBP achieve ReplayFrac=1 with ExactFrac=0; selective release (HARP) meets $c>0$ without a ledger of non-raw scores.

b) How to read Acc / ExactFrac / LiRA.: **Acc** is task utility (primary objective under the SLA). **ExactFrac** is the service constraint: fraction of nodes released as raw $\hat{p}_v$. **LiRA** is an optional audit dial—used to decide how much Frac to spend, not the headline win condition. In practice: if ExactFrac $\geq c>0$ is required, discard rows with ExactFrac=0 even if their LiRA looks better; if only Acc s.t. LiRA $\leq \tau$ matters, prefer equal-mass LBP and ignore HARP. Our running Cora example ($c=0.60$, $n_{\text{sh}}=16$): equal-mass LBP Acc ≈ 0.87 /LiRA ≈ 0.59 /ExactFrac=0 vs. HARP Acc ≈ 0.83 /LiRA ≈ 0.67 /ExactFrac=0.60 with much lower clean-majority ECE.

c) Framework.: **HARP** is a constrained score-serving layer for GNN APIs: (1) a *constructor* ranks nodes (topology, degree, train-neighbor exposure, confidence/entropy, random, or audit-guided vulnerability); (2) hop expansion $\mathcal{P}_k = \text{Ball}_k(\mathcal{R})$ plus a *protector* only on $\mathcal{P}_k$; (3) operator dials Frac/Mass/session budget B , with **CFS** searching Frac under (c, τ) . Default: *release-only* selective Laplace at Frac=0.40 (ExactFrac=0.60).

d) Headline evidence.: At $n_{\text{shadows}}=16$, HARP meets ExactFrac ≥ 0.60 while uniform and stateful-noisy LBP cannot; equal-mass LBP remains the right comparator when $c=0$ is allowed. Within the constrained class, constructors cut clean-slice AUROC ($0.669 \rightarrow 0.594$), CFS selects Frac under operator (c, τ) , and HARP $\circ$ GAP preserves GAP’s ε for public-metadata constructors. ogbn-arxiv/products probes show Acc and serving cost at volume; privacy audits at that scale are scoped and canary-stressed (not claimed as full-graph LiRA).

B. Gap

Prior release defenses (LBP [5], MemGuard [6]) perturb every response and therefore yield $\text{ExactFrac}=0$ unless wrapped in a response ledger that stores non-raw scores. Train-time GNN MIA methods (GTD, MaskArmor) [21], [22] and DP-GNNs [13], [14], [36] do not expose $\text{ExactFrac}/\text{Mass}$ serving accounting for prediction APIs. None formalize Acc under $\text{ExactFrac} \geq c > 0$ as a Big-Data score-serving objective with constructors, validation-time Frac search, and trainer composition.

C. Contributions

- 1) **ExactFrac-constrained API release.** Separate RawExactFrac from ReplayFrac for prediction serving; show fresh and stateful-noisy uniform policies fail $\text{ExactFrac} \geq c > 0$; hop coverage and DP post-processing composition (Prop. 2).
- 2) **HARP serving layer.** Selective, hop-aware release with constructors, validation-time CFS, optional DCS, and $\text{HARP} \circ \text{GAP}$ for finite ε plus ExactFrac controls under continuous query traffic.
- 3) **Constraint-aware Big-Data evaluation.** Headline LiRA at $n_{\text{sh}}=16$; RawExactFrac -vs- ReplayFrac SLA (Table VIII); cache microbenchmarks; scoped OGB $\text{Acc}/\text{serving}$ and canary-stressed probes; clean-slice risk as a structural limit.

II. BACKGROUND AND RELATED WORK

Shadow/confidence MIAs [1]–[3] and LiRA [4] are the audit standard for prediction APIs; overconfidence is a recurring cue [17]. On graphs, Olatunji et al. [5] introduce node-level MIAs and LBP; He et al. [38], GTD [21], MaskArmor [22], and surveys [8] study related attacks/defenses. MemGuard [6] adversarially perturbs every confidence vector ($\text{ExactFrac}=0$). DP-GNNs such as GAP [13], [14] provide train-time (ε, δ) ; structure modulates leakage [18], [39]. **Positioning:** This is a prediction-API / score-serving paper with an optional membership audit dial. LBP/MemGuard spend on every node ($\text{ExactFrac}=0$ unless a non-raw ledger is added); GTD/MaskArmor harden training without ExactFrac accounting; GAP answers formal ε ($\text{ExactFrac}=1$ at eval for train-only noise); HARP answers $\text{ExactFrac} \geq c$ serving under continuous query traffic and composes with DP trainers under Prop. 2. Our GAP-agg and MemGuard baselines are faithful reimplementations, not line-by-line ports of the official codebases—we label them as such throughout. Primary comparisons under $c > 0$ are selective policies at matched Frac ; equal-mass LBP is the unconstrained ($c=0$) reference.

III. PROBLEM SETUP

A. Notation

Let $G = (V, E)$ be an undirected graph with $n = |V|$ nodes, feature matrix $X \in \mathbb{R}^{n \times d}$, and labels $y \in \{0, \dots, C-1\}^n$. A random split partitions V into supervised train $\mathcal{T}$, validation $\mathcal{V}$, and test $\mathcal{Q}$ with ratios 40%/20%/40% (resampled per seed). Training is *transductive*: every node participates in message

passing; only $\mathcal{T}$ enters the cross-entropy loss. We write $\hat{p}_v$ for the softmax posterior at v , r_v for a risk score in $[0, 1]$, $\mathcal{P}_k$ for the hop-expanded protected set, $\text{Mass} = \sum_v \sigma_v$, and $\text{Frac} = |\mathcal{P}_k|/n$.

B. Constrained score release

Operators rarely optimize Acc alone. They may need usable accuracy *and* a positive fraction of unmodified model posteriors, subject to a membership audit.

Definition 1 (Constrained Score Release). *Given a trained GNN and release channel, choose a release policy Π to*

$$\max_{\Pi} \text{Acc}(\Pi) \quad \text{s.t.} \quad \text{LiRA}(\Pi) \leq \tau, \quad \text{ExactFrac}(\Pi) \geq c,$$

where $\text{ExactFrac} \equiv \text{RawExactFrac} = \Pr_v[\tilde{p}_v = \hat{p}_v]$ (fraction of nodes released as the unmodified posterior), $\text{ReplayFrac} = \Pr_v[\tilde{p}_v^{(1)} = \tilde{p}_v^{(2)}]$ is reported separately, τ is an operator-chosen audit risk tolerance matched to the LiRA budget (not a universal privacy target), and $c \in [0, 1]$ is an operator-chosen clean fraction (study default 0.60). Under release-only selective Laplace with $\text{weak}=0$, $\text{ExactFrac}=1 - \text{Frac}$ by construction and implies ReplayFrac on the clean slice without a response ledger.

Remark 1 (Uniform and stateful-noisy LBP under $c > 0$). *Independent Laplace noise with scale $b > 0$ on every coordinate changes $\hat{p}_v$ almost surely, so $\text{ExactFrac}=0$ for any uniform fresh LBP policy. A global per-node response cache or a deterministic PRNG keyed by (node, epoch) can raise ReplayFrac to 1 and restore cache hits / ledger audit, but ExactFrac remains 0 (Table VIII). Selective release with protected fraction Frac achieves $\text{ExactFrac}=1 - \text{Frac}$ by construction, without storing non-raw scores.*

This reframes the Acc –LiRA comparison. Equal-mass LBP can win Acc subject to $\text{LiRA} \leq \tau$ when $c=0$ is allowed; HARP solves the strictly harder problem with $\text{ExactFrac} \geq c > 0$. The Frac sweep traces HARP’s Pareto frontier under that constraint.

C. Threat model

The adversary is a black-box client: it queries softmax scores and tries to guess whether a node was in the training set [2], [4], [5]. It does not see gradients or embeddings. White-box parameter access remains out of scope; label-only gap AUROC and edge-similarity membership AUROC are reported alongside LiRA (GAP-agg cuts edge AUROC $0.895 \rightarrow 0.748$ on Cora).

We also stress three practical variants: defense-aware shadows, multi-query averaging ($K \in \{1, 5, 20\}$), and an LTE-aware query adversary that focuses on clean nodes, protected-set boundaries, and top-decile LTE seeds (Sec. VI-B01). We do not claim robustness to poisoning or unbounded queries without a session budget.

HARP: ExactFrac-constrained selective score serving

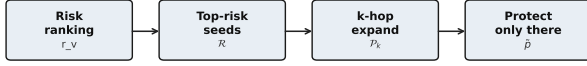


Fig. 1. HARP pipeline: risk ranking $\rightarrow$ high-risk seeds $\rightarrow k$ -hop expansion $\rightarrow$ protector only on the protected set (ExactFrac=1 – Frac on the clean majority).

D. Metrics

Utility: test Acc (primary). **Audit:** LiRA AUROC [4] and TPR@1% FPR, plus Salem-style ϕ /confidence attacks [2]. **Fidelity:** ECE [17], clean-slice ECE, ExactFrac. LiRA decides *whether* to spend noise; Acc/ECE measure the cost.

Remark 2. *The primary protocol is transductive (non-members remain in the graph during training), matching common semi-supervised GNN APIs. We additionally report an inductive edge-exclusion check on Cora (Sec. VI-B0m) where test-incident edges are dropped at train/infer time.*

IV. HARP: HOP-AWARE SELECTIVE RELEASE

A. Design principle

Uniform *fresh* posterior noise cannot meet ExactFrac $\geq c > 0$ —any independent $b > 0$ draw changes $\hat{p}_v$. Stateful wrappers (global caches, ledgers, seeded PRNGs) can meet ReplayFrac without ExactFrac (Table VIII). Node-local selection without hop coverage is incomplete for GNNs: if v is protected but neighbor u is released exact, aggregation can reintroduce membership signal into u ’s score. HARP is a three-interface serving framework (Fig. 1):

- 1) **Constructor (pluggable):** rank nodes—topology (LTE), degree, train-neighbor exposure, confidence/entropy, random, or audit-guided vulnerability.
- 2) **Geometry + protector:** expand $\mathcal{P}_k = \text{Ball}_k(\mathcal{R})$; apply Laplace or masking only on $\mathcal{P}_k$.
- 3) **CFS + dials:** validation-time search of Frac under ExactFrac $\geq c$ and optional LiRA $\leq \tau$; log Mass/Frac/ B .

Locked study config: *release-only* Laplace, Frac=0.40 (ExactFrac=0.60), $k=1$, $\sigma=0.30$. Topology LTE is the no-shadow default constructor; ensemble/audit-guided ranking is preferred when shadows are already paid for. Train-time alignment is an optional ablation—not required for the ExactFrac claim.

a) *Noise mass and clean fraction.*: Mass = $\sum_v \sigma_v$ and Frac = $|\mathcal{P}_k|/n$ are operator accounting metrics. Uniform LBP at scale b has Mass= bn and Frac=1. See paragraph IV-E0a for the locked-budget identity.

b) *Locked-config example (Cora).*: On Cora ($n=2708$), Frac=0.40 protects ≈ 1083 nodes after one-hop expansion; ≈ 1625 nodes stay exact. Mass_{HARP} ≈ 325 matches equal-mass LBP with $b=0.12$. Strong LBP ($b=0.3$) spends Mass ≈ 812 on all nodes—the Acc-collapse baseline in Table II.

c) *Release transform (Laplace protector).*: Let $\sigma_v = \sigma_{\text{strong}} \mathbb{1}[v \in \mathcal{P}_k]$. The released posterior is

$$\tilde{p}_v = \text{renorm}([\hat{p}_v + \text{Lap}(0, \sigma_v)]_+), \quad (1)$$

with independent Laplace coordinates and the same transform on every defense-aware shadow. The masking protector instead returns $\hat{p}_v$ for $v \notin \mathcal{P}_k$ and a one-hot of $\arg \max \hat{p}_v$ for $v \in \mathcal{P}_k$.

d) *Algorithm (systems view).*:

- 1) **Risk:** $r \leftarrow \text{Constructor}(G, \mathcal{T})$ (topology / degree / confidence / random / audit).
- 2) **CFS (optional):** search Frac $\leq 1 - c$ to maximize Acc subject to LiRA $\leq \tau$.
- 3) **Seeds + expand:** binary-search seeds so $|\text{Ball}_k(\mathcal{R})|/n \approx \text{Frac}$; $\mathcal{P}_k \leftarrow \text{Ball}_k(\mathcal{R})$.
- 4) **Release:** apply (1) (or masking); log (Mass, Frac, Acc, LiRA, ECE).

Train-time alignment is optional and omitted in the locked release-only configuration.

B. Risk constructors

The constructor interface is part of the framework contribution. **Topology (LTE)** uses only $G, y|_{\mathcal{T}}$, and the train mask (inverse degree, local heterophily, train-neighbor exposure); it is a cheap default when shadows are unavailable—not claimed to be privacy-optimal. **Degree / train-neighbor** isolate individual topology cues. **Confidence / entropy** rank from a clean forward pass (high max-confidence or high entropy). **Random** is the matched-Frac selectivity control. **Audit-guided** scores $|\mu_{\text{IN}} - \mu_{\text{OUT}}|$ over a small shadow ensemble [37] and is preferred when that ensemble is already paid for. We report Spearman correlation between each constructor and audit vulnerability, plus Acc/LiRA/slice AUROC at matched Frac (Table VII). On Cora/Citeseer, *entropy* is the strongest cheap prior (Spearman ≈ 0.33 – 0.35); topology (LTE) is moderate (≈ 0.25 – 0.28); max-confidence is anti-correlated. Heterophilic Chameleon priors are weak—selectivity still enforces ExactFrac, but ranking gains are limited. Where topology underperforms random on LiRA (e.g., Cora in Table VII), we state that plainly: ExactFrac comes from selectivity, not from LTE.

C. Why k -hop expansion

Let $\mathcal{R}$ be seeds. A GNN score at u already mixes information from $\text{Ball}_k(u)$ before any release noise is applied. If a membership-sensitive seed $v \in \mathcal{R}$ is noised at release but a neighbor u is returned bit-exact, the adversary simply queries u .

Proposition 1 (Hop coverage under a one-hop linear mixer). *Fix a one-hop score model $s_u = \alpha z_v + (1 - \alpha) z_u$ for $u \in \mathcal{N}(v)$, $\alpha \in (0, 1]$, where z_v carries a membership cue of gap $\gamma > 0$, and release adds independent Laplace noise only to chosen nodes’ scores. If the policy noises v but leaves u exact, an adversary observing s_u retains the unprotected leaf advantage. Noising $\text{Ball}_1(\{v\})$ places Laplace noise on every score that linearly depends on z_v under this mixer.*

Proof sketch. Under seed-only protection the released leaf equals the clean mixer output, so the member/non-member gap in $\mathbb{E}[s_u]$ remains $\alpha\gamma$. Expanding protection to u subjects that coordinate to $\text{Lap}(0, \sigma)$. A synthetic mixer confirms the ordering: unprotected and seed-only leaf AUROCs stay near 1.0, while hop-1 protection lowers the leaf attack to ≈ 0.72 . $\square$

This is a motivating linearized argument for hop expansion, not a necessity theorem for nonlinear multi-layer GNNs. Empirically we set k to the model’s receptive field (default $k=1$ for 2-layer SAGE/GCN; $k \in \{0, 2\}$ changes Acc by $\lesssim 0.01$ at matched Frac). Frac—not k —is the population Acc–Mass knob.

D. Guarantees: CFS configuration and DP composition

Remark 3 (CFS is validation-time configuration, not a novel optimum). *Constrained Frac Search (CFS) enumerates a finite grid $\mathcal{F} \subseteq [0, 1 - c]$ and returns $f^* = \arg \max\{\text{Acc}(f_i) : \text{LiRA}(f_i) \leq \tau\}$ using validation Acc and a validation-time defense-aware LiRA estimate; the locked tables then report a held-out test Acc / headline LiRA once. This is a reproducible operator procedure for choosing Frac under explicit (c, τ) —not a claim of global optimality over all release policies or constructors. The audit cap τ is an operator risk tolerance tied to the attack budget (study headline: $n_{\text{sh}}=16$, $\tau=0.70$); $\tau=0.55$ is infeasible at that budget, and we report neighboring caps $\{0.68, 0.70, 0.75\}$ (Sec. VI-B).*

Proposition 2 (Post-processing composition). *Let T be any training procedure whose released model satisfies (ε, δ) -DP w.r.t. the training data, and let Π be a HARP release policy that is a (possibly randomized) function only of the model’s released outputs and public deployment metadata. Then $\text{HARP} \circ T$ satisfies the same (ε, δ) -DP (standard post-processing [27]) and attains $\text{ExactFrac}=1 - \text{Frac}$ w.r.t. the DP model’s posteriors.*

Proof. Immediate from the post-processing property whenever Π does not re-read private training labels or private membership bits beyond what T already released. $\square$

Prop. 2 is not a new DP theorem; it is the reason HARP can sit *above* a DP trainer as a serving layer. **Constructor caveat:** topology LTE uses the train mask and $y|_{\mathcal{T}}$ as server-side deployment metadata. If those bits are private under the DP threat model, use a public-metadata constructor (degree, confidence/entropy from the released DP outputs, random, or audit scores computed only from released shadow outputs) before claiming Prop. 2. Section VI-B measures $\text{HARP} \circ \text{GAP}$ under the public-metadata reading.

E. Complexity and accounting

Proposition 3 (HARP setup and release complexity). *Let $G = (V, E)$ have $n = |V|$ nodes and $m = |E|$ adjacency endpoints, with C classes. Fix hop radius $k = O(1)$ and $T = O(1)$ binary-search iterations for a target protected fraction. Then setup (LTE + seeds + k -hop expand) runs in $O(m + n \log n)$*

time and $O(n + m)$ space. Release runs in $O(nC)$ —matching uniform LBP’s tier, with less noise-sampling work whenever $\text{Frac} < 1$.

Proof sketch. LTE scans edges then normalizes n scores: $O(m + n)$. Seed selection is $O(n \log n)$ plus T BFS expansions at $O(m + n)$. Release writes scales, samples Laplace on $\mathcal{P}_k$, and renormalizes: $O(nC)$. $\square$

Remark 4 (Per-response Laplace; no global DP claim). *Interpret each node’s clean posterior as a query with ℓ_1 -sensitivity at most $\Delta=2$ on the simplex. Independent $\text{Lap}(\sigma)$ on coordinates of $v \in \mathcal{P}_k$ is a per-response Laplace mechanism with scale σ ; unprotected nodes receive no noise. Any policy with $\text{ExactFrac} > 0$ has infinite global (ε, δ) -DP by construction. We report this only as an accounting remark—not as a local-DP guarantee under graph adjacency—and recommend training-time DP (e.g., GAP) when a legal global ε is required [13].*

a) *Accounting identity (stated once).*: Under locked Frac and fixed σ_{strong} , $\text{Mass}_{\text{HARP}} = \text{Frac} \cdot n \cdot \sigma_{\text{strong}}$ by construction. Mass is an operational dial correlated with mean ℓ_1 distortion and top-1 flip rate (at matched Mass 325 on Cora: HARP mean ℓ_1 0.349 vs. LBP 0.493); it is not a standard privacy budget. Empirical claims are (i) feasibility under $\text{ExactFrac} \geq c$, (ii) Acc/ECE within the selective class at matched Frac/Mass, and (iii) serving telemetry.

F. What HARP is not

HARP is not differential privacy and is not a claim of lower population LiRA than uniform noise under $\text{ExactFrac} = 0$. It is not a claim that randomization “breaks caching”—stateful caches and seeded noise can stabilize non-raw responses. It is a constrained ExactFrac serving framework: spend a protector on a hop-expanded subset, keep a raw-posterior majority, configure Frac with CFS, and audit residual risk—including on unprotected nodes.

V. EXPERIMENTAL PROTOCOL

A. Pre-declared rules

Primary objective: Acc under $\text{ExactFrac} \geq c$ (study default $c=0.60$; swept in Fig. 2). **Primary metrics:** Acc, ExactFrac , ECE (fidelity). **Audit metric:** LiRA AUROC (and clean-slice confidence AUROC) as deployment constraints—not the headline win condition. **Operational metrics:** Mass, Frac, ExactFrac (Raw ExactFrac), ReplayFrac , shared-cache hit rate (Table VIII). Hyperparameters are not selected to minimize test LiRA. Under $c > 0$, primary baselines are selective policies at matched Frac; equal-mass LBP is the unconstrained ($c=0$) reference; sticky/global-cache/seeded LBP are ReplayFrac -positive but ExactFrac -infeasible controls.

B. Systems metrics (definitions)

- **ExactFrac** ($\equiv \text{RawExactFrac}$): fraction of nodes with $\hat{p}_v = \hat{p}_v$.
- **ReplayFrac**: fraction of nodes with equal responses under re-query (may use cache/seed state).

- **Mass** = $\sum_v \sigma_v$: operational Laplace-scale spend (correlated with distortion).
- **Frac** = $|\mathcal{P}_k|/n$: protected fraction (ExactFrac=1 – Frac under weak=0).
- **ECE**: calibration of released probabilities.

C. Locked HARP (release-only)

Release-only selective Laplace: $\lambda=0$, no HCAG, no train-on-protected; $k=1$, $\sigma_{\text{strong}}=0.30$, Frac=0.40 (ExactFrac=0.60). Constructor default in locked tables: topology (LTE); audit/random/degree/confidence appear in Table VII. Train-time alignment is reported only as an ablation. Baselines: equal-mass LBP $b=0.12$; strong LBP $b=0.3$; MemGuard; GTD; MaskArmor; selective masking; DP-SGD Acc-tuned (vacuous ε).

D. Baselines and attacks

Gaussian LiRA with defense-aware shadows: $n_{\text{shadows}}=16$ for the headline Cora table and CFS (Tables II, V; $\tau=0.70$), $n_{\text{shadows}}=4$ for multi-dataset grids and Acc dials, and sweeps $\{4, 16, 64\}$ on Cora/Chameleon. Five seeds with mean $\pm 95\%$ CI on primary Cora/Chameleon/Citeseer constructor tables; paired seeds across defenses. ogbn-arxiv: NeighborLoader serving Acc with LiRA at $n_{\text{shadows}} \in \{2, 4, 16\}$ (audit stress, not a million-node privacy claim). ogbn-products: full-graph serving RSS/QPS; BFS LiRA ladder at $n \in \{15k, 40k\}$, $n_{\text{sh}} \in \{2, 4, 8\}$, plus a canary-stressed 15k probe with undefended canary-LiRA ≈ 1.0 .

E. What each baseline tests

Equal-mass LBP ($b=0.12$) matches HARP Mass at Frac=0.40 and is the unconstrained ($c=0$) reference. **Selective constructors** (random/degree/confidence/entropy/audit/topology) isolate ranking quality at matched Frac under the SLA. **Strong LBP** ($b=0.3$) is a secondary published recipe that collapses Acc under full-graph strong noise. **MemGuard** (reimplementation) fits a standardized MLP ϕ -attack on holdout members/non-members and searches an L_1 ball to drive $P(\text{member}) \rightarrow 0.5$ while preserving argmax (ExactFrac=0). Following Aerni et al. [37] we report both ϕ and LiRA; at $n_{\text{sh}}=16$ our port cuts LiRA vs. undefended ($0.813 \rightarrow 0.667$) while remaining ExactFrac-infeasible. **GAP-agg** is a GAP-inspired aggregation-Laplace baseline: noise is train-only, so eval ExactFrac=1 with analytical (ε, δ) (hybrid DP-train + ExactFrac-serve). Not a full GAP codebase reproduction. **Selective masking** is an alternate protector (ExactFrac preserved; often worse LiRA). **HARP-full** (Frac=1) is the unconstrained ablation that recovers strong-LBP Acc–LiRA. **GTD/MaskArmor** are train-time defenses (no release Mass/Frac). **Clip+noise DP-SGD** remains an Acc-tuned vacuous- ε reference.

F. Attack protocol (summary)

We train defense-aware shadow models, release transformed posteriors, and report LiRA AUROC with ϕ -features [2], [4].

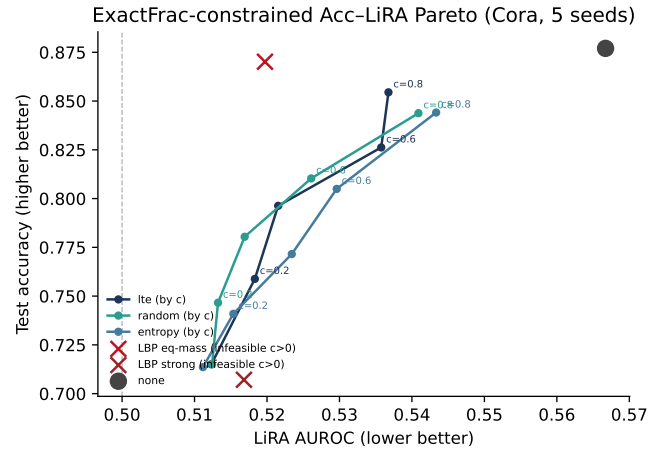


Fig. 2. ExactFrac-constrained Acc–LiRA Pareto on Cora (5 seeds). Selective HARP constructors sweep $c \in \{0.2, 0.4, 0.6, 0.8\}$ (ExactFrac= c). Uniform LBP markers are ExactFrac=0 and infeasible for every $c>0$.

Validation nodes are excluded from membership labels; test Acc/ECE use the held-out test split. Multi-query experiments average $K \in \{1, 5, 20\}$ independent draws; session policies cap fresh draws at $B=5$.

a) *Negative controls.*: PubMed (near-chance undefended LiRA): HARP still preserves Acc vs. strong LBP at Frac=0.40 (0.839 vs. 0.768); ogbn-arxiv/products give near-chance or memory-bounded audits under matched budgets—audit first, harden second [28].

VI. RESULTS

A. Overview

a) *Operational headline (constraint-first).*: Under ExactFrac ≥ 0.60 : (i) HARP is feasible and uniform/stateful-noisy LBP are not (Table I); (ii) release-only HARP keeps Acc competitive within the selective class while returning a 60% raw-posterior majority; (iii) cache microbenchmarks show hit-rate gains when many responses are bit-stable (Table IX); (iv) equal-mass LBP can still win Acc–LiRA when $c=0$ is allowed—that unconstrained privacy-only regime is out of scope for the SLA.

b) *Reading the tables.*: Compare policies only inside the feasible ExactFrac set for your c . Acc is the primary utility number; LiRA/TPR@1% FPR tell the operator whether to raise Frac or tighten τ ; ECE reports score quality on the released (especially clean) majority. Questions: **(Q1)** feasibility under $c>0$; **(Q2)** Acc/ECE vs. selective and equal-mass baselines; **(Q3)** constructors vs. random on utility and slice privacy; **(Q4)** serving cost and clean-slice residual risk; **(Q5)** OGB volume Acc/serving with scoped audits.

B. HARP under ExactFrac ≥ 0.60 : accuracy and score quality

a) *Primary claim.*: Fig. 3, Fig. 4, and Table II: under ExactFrac ≥ 0.60 , release-only HARP is feasible; equal-mass LBP and MemGuard are not. At $n_{\text{sh}}=16$ the undefended attack is strong (LiRA 0.813, TPR@1% 0.200); HARP cuts it to

TABLE I
FEASIBILITY UNDER $\text{EXACTFRAC} \geq c$ (RAW EXACTFRAC).
STICKY/GLOBAL/SEEDED LBP CAN RAISE REPLAYFRAC BUT REMAIN
 EXACTFRAC -INFEASIBLE FOR $c > 0$.

Policy	$c=0$	$c=0.2$	$c=0.6$	$c=0.8$
none	yes	yes	yes	yes
LBP fresh	yes	no	no	no
LBP sticky / global / seeded	yes*	no†	no†	no†
HARP (Frac=1 - c)	yes	yes	yes	yes

*ReplayFrac can be 1 (Table VIII). †ExactFrac=0: responses are never the unmodified $\hat{p}_v$.

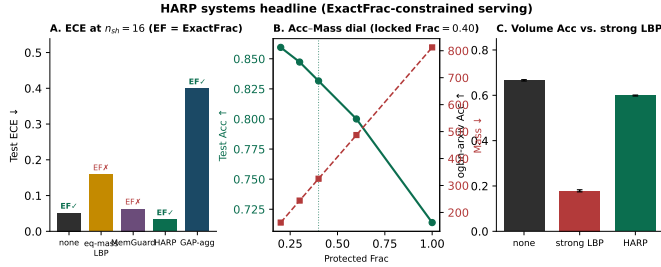


Fig. 3. Systems headline for ExactFrac-constrained serving. (A) ECE at $n_{sh}=16$ with ExactFrac feasibility marks ($\checkmark/\times$). (B) Frac Acc-Mass dial (locked Frac=0.40). (C) ogbn-arxiv Acc vs. strong LBP (volume Acc, not a full-graph privacy claim).

TABLE II
CORA GRAPHSAGE UNDER $\text{EXACTFRAC} \geq c=0.60$ (5 SEEDS; LiRA $n_{sh}=16$). FEASIBLE UNDER $c > 0$: NONE, GAP-AGG, HARP.
†GAP-INSPIRED AGG. LAPLACE (ANALYTICAL ε ; NOT FULL GAP CODE).
‡SHADOW-FIT MEMGUARD; ϕ = ATTACK IT OPTIMIZES. ¶HARP
FRAC=1 RECOVERS STRONG-LBP ACC-LiRA.

Defense	Feas.	Acc	LiRA	TPR@1%	ϕ	ECE↓	EF
none	yes	0.877	0.813	0.200	0.650	0.051	1.00
GAP-agg ($\sigma=3$, $\varepsilon \approx 41$)†	yes	0.776	0.624	0.039	0.540	0.399	1.00
HARP (release)	yes	0.826	0.674	0.060	0.626	0.033	0.60
LBP equal-mass	no	0.870	0.592	0.009	0.584	0.159	0.00
MemGuard‡	no	0.877	0.667	0.080	0.619	0.063	0.00
LBP $b=0.3$ / HARP-full¶	no	0.707 / 0.715	0.551 / 0.560	0.007 / 0.008	0.565 / 0.572	0.142 / 0.152	0.00

0.674 / 0.060 while keeping Acc 0.826, ECE 0.033, and a 60% replay-stable majority. MemGuard is a credible unconstrained baseline (LiRA 0.667) but ExactFrac=0; equal-mass LBP reaches LiRA 0.592 with ECE 0.159 and ExactFrac=0. Equal-mass can win Acc-LiRA when the SLA is dropped—expected and outside the constrained objective.

b) *Secondary: Acc vs. published strong LBP.*: Against LBP $b=0.3$ [5], HARP raises Acc 0.707 $\rightarrow$ 0.832 at Frac=0.40 because strong noise applies to a minority of nodes (Table II). This is mechanically expected and is not the constrained-feasibility claim.

c) *Membership audit (deployment dial).*: At matched Mass, equal-mass LBP beats HARP on Acc-LiRA (Cora $n_{sh}=16$: Acc 0.870 / LiRA 0.592 vs. 0.826 / 0.674). If ExactFrac is not required, prefer equal-mass LBP. Under the SLA, LiRA guides whether Frac must rise and whether the audit threshold τ is met after CFS.

HARP Mass 325; LBP eq-mass 325; strong LBP 812. MemGuard is a credible unconstrained baseline (LiRA 0.813 $\rightarrow$ 0.667) but ExactFrac-infeasible; HARP is within ≈ 0.01 LiRA of MemGuard with ExactFrac=0.60 and much lower ECE. At

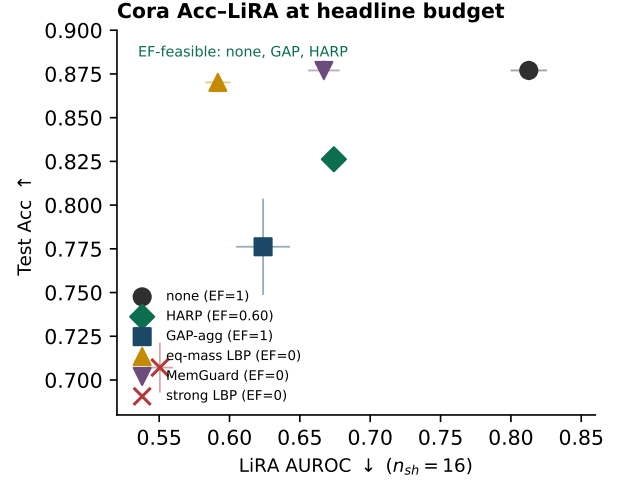


Fig. 4. Cora Acc-LiRA at the headline budget ($n_{sh}=16$, 5 seeds). ExactFrac-feasible: none, GAP-agg, HARP. Equal-mass LBP/MemGuard win unconstrained Acc-LiRA but ExactFrac=0.

$n_{sh}=4$: GTD 0.878/0.538; MaskArmor 0.877/0.580.

GTD/MaskArmor report Acc/LiRA/ECE only (train-time; no Mass/Frac). GAP-agg is ExactFrac-feasible (train-only noise $\Rightarrow$ ExactFrac=1) with finite ε , at Acc/ECE cost. HARP is the high-Acc ExactFrac option without training DP; HARP-full recovers strong-LBP Acc-LiRA under $c=0$. Selective masking keeps ExactFrac=0.60 but raises LiRA.

d) *HARPoGAP: DP trainer as a component.*: Proposition 2 says selective release is post-processing, so composing HARP on a GAP-trained model preserves GAP's (ε, δ). Measured (Cora, 5 seeds, $\sigma=3$): GAP alone Acc 0.770 ± 0.030 , LiRA 0.520; HARPoGAP at Frac=0.40 keeps the same analytical $\varepsilon \approx 41$, ExactFrac=0.60, LiRA 0.517, clean-slice conf AUROC 0.584, at Acc 0.649—composition costs Acc because GAP posteriors already carry training noise. The large ε is the Acc-preserving operating point of our GAP-agg reimplementation ($\sigma=5$ yields $\varepsilon \approx 19$ at Acc 0.44); we do not market $\varepsilon \approx 41$ as a strong privacy guarantee—only as a finite analytical budget that HARP does not inflate. Operators who need a formal ε and selective-release controls can compose; operators who need only ExactFrac keep plain HARP's higher Acc.

Slice ECE localizes the win on Cora (5 seeds): $\text{ECE}_{\text{clean}}=0.018$ vs. $\text{ECE}_{\text{prot}}=0.081$ (population 0.033); clean nodes match undefended calibration, protected nodes carry the noise penalty. The pattern is portable: clean-slice ECE beats undefended population ECE on Cora/Citeseer/Chameleon (0.020/0.083/0.273 vs. 0.053/0.140/0.290), while equal-mass LBP perturbs every response even when its population ECE looks competitive (Citeseer 0.073). ΔAcc vs. strong LBP at Frac=0.40 ranges from +0.040 (Actor) to +0.185 (Photo) across six graphs at matched telemetry ($\text{Mass}_{\text{HARP}}=0.4 \text{ Mass}_{\text{LBP}}$).

e) *Multi-dataset portability.*: Table III checks that selective release remains ExactFrac-feasible across six graphs under

TABLE III

HARP PORTABILITY GRID (GRAPH-SAGE; CORA 5-SEED, OTHERS 3-SEED; $n_{sh}=4$). HEADLINE CORA AUDITS USE $n_{sh}=16$ (TABLE II). TPR = LiRA TPR@1% FPR.

Dataset	Defense	Acc	LiRA	TPR	Mass	Frac
Cora	none	0.877	0.567	0.033	—	—
	LBP $b=0.3$	0.707	0.517	0.019	812	1.00
	HARP	0.832	0.533	0.011	325	0.40
Citeseer	none	0.744	0.591	0.042	—	—
	LBP $b=0.3$	0.620	0.523	0.030	998	1.00
	HARP	0.716	0.548	0.014	399	0.40
Chameleon	none	0.448	0.618	0.059	—	—
	LBP $b=0.3$	0.391	0.539	0.038	683	1.00
	HARP	0.452	0.555	0.011	273	0.40
Actor	none	0.335	0.610	0.037	—	—
	LBP $b=0.3$	0.287	0.549	0.032	2280	1.00
	HARP	0.327	0.561	0.019	912	0.40
PubMed	none	0.876	0.505	0.012	—	—
	LBP $b=0.3$	0.768	0.499	0.010	5915	1.00
	HARP	0.839	0.502	0.012	2366	0.40
Photo	none	0.802	0.500	—	—	—
	LBP $b=0.3$	0.552	0.499	—	2295	1.00
	HARP	0.737	0.501	—	946	0.40

TABLE IV

HARP FRAC SWEEP ON CORA (5 SEEDS; $\sigma_{strong}=0.30$). ACC/MASS ARE BUDGET-INVARIANT; LiRA COLUMN IS $n_{sh}=4$ (SEE TABLE II/ V FOR $n_{sh}=16$).

Frac	Acc $\uparrow$	LiRA $\downarrow$	Mass $\downarrow$
0.20	0.859 [0.855, 0.866]	0.537 [0.532, 0.543]	162
0.30	0.847 [0.843, 0.853]	0.538 [0.533, 0.543]	244
0.40	0.832 [0.826, 0.839]	0.533 [0.525, 0.541]	325
0.60	0.800 [0.793, 0.809]	0.519 [0.512, 0.526]	487
1.00	0.714 [0.699, 0.727]	0.511 [0.503, 0.519]	812

a matched light audit ($n_{sh}=4$; headline Cora uses $n_{sh}=16$). The secondary comparator is published strong LBP ($b=0.3$), which collapses Acc under full-graph noise; HARP recovers 4–19 Acc points at Frac=0.40 while keeping ExactFrac=0.60. This is *not* an unconstrained Acc–LiRA win (equal-mass LBP remains preferred when $c=0$). Heterophilic rows (Chameleon, Actor) show the largest undefended LiRA and the clearest case for a Frac budget; PubMed/Photo are near-chance controls where operators should set Frac=0 unless ExactFrac itself is required.

f) *Frac dial and Mass accounting.*: Table IV sweeps Frac on Cora (5 seeds): Acc falls monotonically as Frac rises under fixed σ_{strong} —the systems Acc–Mass dial. LiRA in this sweep uses $n_{sh}=4$ (cheap dial diagnostics); the headline privacy comparison is Table II at $n_{sh}=16$, and CFS under that budget is Table V.

g) *Constrained Frac Search (matched to the headline audit).*: CFS is only meaningful when τ tracks the attack budget: $\tau=0.55$ is appropriate under $n_{sh}=4$ but infeasible under headline $n_{sh}=16$ (locked HARP LiRA ≈ 0.67). We treat τ as an operator-selected risk tolerance—not a universal privacy target—and run CFS at $n_{sh}=16$ with ExactFrac ≥ 0.60 and study cap $\tau=0.70$ on *validation* Acc/LiRA (Table V); locked Frac=0.40 remains the pre-declared release-only dial reported on the held-out audit. Among Frac $\in \{0, 0.2, 0.3, 0.4\}$, every constructor finds a feasible point; mean selected Frac is

TABLE V

CFS ON CORA AT $n_{sh}=16$ UNDER EXACTFRAC ≥ 0.60 AND LiRA $\leq \tau=0.70$ (3 SEEDS; MEAN SELECTED FRAC/ACC/LiRA). ALL ROWS FEASIBLE_AUDIT=1.

Constructor	Selected Frac	Acc	LiRA
topology (LTE)	0.33	0.838	0.683
ensemble	0.30	0.838	0.681
random	0.23	0.839	0.689

TABLE VI

SESSION BUDGET ON HARP CORA AT $K=20$ (5 SEEDS). $B=1$ FREEZES ACC NEAR ONE-SHOT; $B=20$ MATCHES UNCAPPED. SHARED GLOBAL $B=1$ ACROSS N IDENTITIES BLOCKS SYBIL ACC RECOVERY (RIGHT).

Policy	B	Acc	LiRA	N ids	Per-id $B=1$	Shared $B=1$
Budget reuse	1	0.832	0.540	1	0.825	0.825
Budget reuse	5	0.851	0.540	5	0.874	0.825
Budget reuse	20	0.885	0.545	20	0.876	0.825
Uncapped avg	—	0.886	0.545	—	—	—

0.23–0.33 with Acc ≈ 0.84 —strictly better Acc than the locked Frac=0.40 dial while meeting τ . Tighter $\tau=0.68$ pushes topology to Frac=0.40; looser $\tau=0.75$ selects Frac=0.20. Under ExactFrac alone (no τ), CFS degenerates to Frac=0; the audit cap is what forces a nonzero spend. Locked Frac=0.40 remains the conservative fixed operating point.

h) *Exact scores $\neq$ membership safety.*: Clean nodes are well calibrated (ECE_{clean}=0.018), but they are not automatically hard to attack. Under locked HARP, unprotected-slice confidence AUROC is ≈ 0.64 vs. ≈ 0.51 population-wide (Table XI); the mitigation stack below addresses it. Aerni et al. [37] motivate such canary checks (Sec. VI, canaries).

i) *Velocity: multi-query and session budgets.*: Averaging K independent releases denoises Laplace and moves toward the undefended baseline (Cora 5-seed LiRA at $K=1/5/20$: HARP 0.536/0.546/0.551; $n_{sh}=4$). A session policy of at most B fresh draws per *authenticated principal*—reusing the last draw thereafter—caps this drift (Table VI). **Enforcement assumption:** server-side state bound to authenticated identities with rate limits. Sybil-capable attackers who mint many principals can bypass per-identity B ; Table VI shows $N=5$ identities under per-id $B=1$ recover Acc ≈ 0.874 (near uncapped), while a *shared global* $B=1$ pool keeps Acc at the one-shot floor (0.825). Deploy shared B (or authenticated single-principal sessions) when Sybil velocity is in scope; Frac and B remain complementary controls.

j) *Constructors: a privacy effect, not just an interface.*: The decisive constructor metric is *clean-slice* conf AUROC: ranking removes vulnerable members from the bit-exact slice (Table VII). At matched ExactFrac=0.60 (5-seed Cora), slice AUROC falls with ranking quality: random 0.669 ± 0.015 , topology 0.647 ± 0.012 , audit 0.616 ± 0.016 , ensemble 0.594 ± 0.010 . Paired tests vs. random: audit $\Delta=0.053$ ($p<10^{-3}$), ensemble $\Delta=0.075$ ($p<10^{-3}$); Acc is equal-or-better (ensemble 0.813 vs. random 0.811). At the headline budget ($n_{sh}=16$, Frac=0.40), ensemble also lowers population LiRA vs. topology (0.646 vs. 0.674)—constructors are pri-

TABLE VII

RELEASE-ONLY CONSTRUCTORS AT FRAC=0.40 (CORa, 5 SEEDS; MEAN $\pm$ 1.96 SE). SLICE = CLEAN-SLICE CONF AUROC. LiRA HERE $n_{sh}=4$; HEADLINE IS TABLE II.

Policy	Feas.	Acc	LiRA	Slice $\downarrow$	EF
none	yes	0.877 $\pm$ 0.005	0.567 $\pm$ 0.006	—	1.00
GAP-agg ($\sigma=3$)	yes	0.776 $\pm$ 0.024	0.518 $\pm$ 0.008	—	1.00
LBP equal-mass	no	0.870 $\pm$ 0.005	0.520 $\pm$ 0.011	—	0.00
LBP $b=0.3$ / HARP-full	no	0.707 / 0.715	0.517 / 0.512	—	0.00
sel. audit	yes	0.820 $\pm$ 0.010	0.526 $\pm$ 0.009	0.616 $\pm$ 0.016	0.60
sel. ensemble	yes	0.817 $\pm$ 0.011	0.527 $\pm$ 0.014	0.594$\pm$0.010	0.60
sel. random	yes	0.810 $\pm$ 0.006	0.526 $\pm$ 0.007	0.669 $\pm$ 0.015	0.60
sel. entropy	yes	0.805 $\pm$ 0.010	0.530 $\pm$ 0.014	—	0.60
sel. topology (LTE)	yes	0.826 $\pm$ 0.006	0.536 $\pm$ 0.008	0.647 $\pm$ 0.012	0.60

TABLE VIII

RAWEXACTFRAC VS REPLAYFRAC (CORa, 3 SEEDS). GLOBAL-CACHE AND SEEDED LBP RESTORE REPLAY/CACHE WITHOUT RAW POSTERIOR EQUALITY; HARP MEETS EXACTFRAC= c WITHOUT A NON-RAW LEDGER.

Policy	RawEF $\uparrow$	Replay $\uparrow$	Cross $\uparrow$	Cache $\uparrow$
none	1.00	1.00	1.00	0.87
LBP fresh	0.00	0.00	0.00	0.00
LBP sticky ($B=1$)	0.00	1.00*	0.00	0.62
LBP global cache	0.00	1.00	1.00	0.87
LBP seeded (node,epoch)	0.00	1.00	1.00	0.87
HARP (Frac 0.40)	0.60	0.60	0.60	0.77

*Within-client sticky replay; cross-client ExactFrac stays 0. Clean-slice threshold flicker $\approx$ 0.09 for fresh/sticky LBP and 0 for global/seeded/HARP.

TABLE IX

IN-PROCESS RESPONSE-CACHE MICROBENCHMARK (CORa; 20K ZIPF QUERIES; CONTENT KEYS ON RELEASED VECTORS). ABSOLUTE QPS/LATENCY ARE NOT END-TO-END API NUMBERS.

Workload	Policy	Hit	QPS	p50 ms	p95 ms
zipf	none	0.937	4.0×10^6	<0.001	<0.001
zipf	lbp_fresh	0.000	9.3×10^3	0.104	0.116
zipf	harp_fresh	0.606	6.2×10^5	<0.001	0.004
zipf	lbp_B1	0.937	1.5×10^5	<0.001	0.101
zipf	harp_B1	0.937	2.2×10^6	<0.001	<0.001

marily a slice dial, with a measurable population effect under strong audits. Hop $k \in \{0, 1, 2\}$ at matched Frac changes Acc by $\lesssim 0.01$ (release-only); selectivity, not train-time alignment, carries ExactFrac.

k) *Why ExactFrac matters (RawExactFrac $\neq$ ReplayFrac).*: Caches and ledgers can stabilize previously sampled noisy outputs [40], [41]; that is ReplayFrac, not ExactFrac. Table VIII (CorA; 3 seeds; 8 clients; 20k Zipf; shared LRU 1024) makes the separation concrete: fresh LBP has both fractions ≈ 0 ; per-client sticky LBP raises within-client ReplayFrac but not cross-client equality or ExactFrac; *global* first-draw caches and deterministic seeded LBP achieve ReplayFrac=1 and high cache hits with ExactFrac=0; HARP alone among noisy policies meets ExactFrac=0.60 without storing non-raw scores. Table X confirms clean-slice bit-stability; Table IX is an *in-process response-cache microbenchmark* (not end-to-end networked API throughput; absolute QPS/latency are mechanism artifacts on CorA).

TABLE X

MEASURED (NOT DEFINITIONAL) REPLAY BEHAVIOR ACROSS INDEPENDENT RE-QUERIES (CORa, 3 SEEDS): BITWISE-EQUAL FRACTION, TOP-1 LABEL FLIPS, AND THRESHOLD ($p_{top} > 0.5$) ROUTE FLIPS. HARP'S FLIPS ARE CONFINED TO THE PROTECTED MINORITY; THE CLEAN 60% IS BIT-STABLE WITH ZERO FLICKER.

Policy	Bit-equal re-query	Top-1 flip	Thresh. flip
none	1.000	0.000	0.000
GAP-agg	1.000	0.000	0.000
LBP eq-mass	0.000	0.027	0.101
HARP (Frac 0.40)	0.600	0.146*	0.181*

*Entirely within the protected 40% ($\sigma=0.30$); clean-slice flip rate is exactly 0. LBP spreads lower-amplitude flicker over *all* nodes, so no response is replay-stable.

TABLE XI

LTE-AWARE ADAPTIVE QUERIES ON CORa HARP (5 SEEDS; CONF AUROC IS ATTACK-BUDGET FREE; LiRA@0.4 USES $n_{sh}=4$).

Query set	Conf@0.4	Conf@0.6	LiRA@0.4
Population	0.518	0.473	0.533
Unprotected (clean)	0.645	0.633	0.542
Boundary (clean)	0.634	0.628	0.538
Top-decile LTE	0.641	0.511	0.529

l) *Clean-slice residual risk (first-class) and the mitigation stack.*: Unprotected nodes are *not* claimed safe (Table XI); clean-slice conf AUROC rises to 0.63–0.67 while population LiRA stays ≈ 0.53 . HARP attacks this from three composable directions (all at ExactFrac=0.60, 5 seeds): (i) *constructor*: ensemble ranking removes vulnerable members from the exact slice, $0.669 \rightarrow 0.594$ (Table VII); (ii) *deterministic confidence smoothing (DCS)*: released rows on the clean slice with $p_{max} > \theta$ are flattened by a fixed temperature ($\theta=0.9$, $T=3$)—a deterministic, argmax-preserving post-map, so re-queries stay bitwise stable, caches and audit replay still work, and routing decisions are unchanged; slice AUROC falls $0.650 \rightarrow 0.618$ under topology ranking and $0.591 \rightarrow 0.576$ on top of ensemble, with *zero* Acc change; (iii) *slice-aware CFS*: adds a slice-AUROC cap to the search. DCS trades raw-score fidelity on smoothed rows for slice privacy while preserving ReplayFrac and threshold stability; when ExactFrac (raw $\hat{p}_v$) is contractual, skip DCS. The stacked floor (≈ 0.575) is structural at $c=0.60$: a 60% deterministic slice must retain some membership signal. Lower c , or use GAP-agg / HARP $\circ$ GAP, if tighter slice control is required.

m) *Inductive edge exclusion.*: Dropping test-incident edges (CorA, 5 seeds) raises undefended LiRA to 0.818; HARP recovers Acc $0.476 \rightarrow 0.601$ vs. strong LBP at Frac=0.40.

C. When to spend: volume $\times$ structure

Table XII(a): undefended LiRA concentrates in small/mid heterophilic cells; Photo/PubMed/ogbn-arxiv sit near chance under matched shadows—audit first, then set Frac. Table XII(b) reports the honest shadow-count story on CorA for none / LBP / HARP. Absolute LiRA rises with $n_{shadows}$ for every defense [4]; Acc is invariant. HARP does *not* beat strong LBP on LiRA at large shadow counts (CorA $n_{sh}=64$: LBP 0.561 vs. HARP 0.677)—expected when ExactFrac=0

TABLE XII

AUDIT CONTROLS (GRAPHSAGE; 3 SEEDS). *HETEROPHILIC.
(B) SHADOW SCALING (NOT HARP-ONLY). ACC INVARIANT IN (B): NONE
0.879, LBP 0.716, HARP 0.836.

(a) Undef. LiRA ($n_{sh}=4$)				(b) Cora LiRA / TPR@1%			
Dataset	n	Acc	LiRA	n_{sh}	none	LBP	HARP
Chameleon*	2277	0.449	0.623	4	0.568/0.033	0.517/0.022	0.534/0.014
Cora	2708	0.879	0.568	16	0.808/0.201	0.555/0.008	0.670/0.057
Actor*	7600	0.335	0.610	64	0.847/0.265	0.561/0.008	0.677/0.075
Photo	7650	0.825	0.500				
PubMed	19717	0.876	0.505				
ogbn-arxiv	169343	0.666	0.503				

is allowed. What HARP keeps is (i) strictly lower LiRA than undefended and (ii) Acc recovery vs. strong LBP ($\Delta\text{Acc} +0.12$, invariant in n_{sh}). TPR@1% FPR follows the same pattern.

D. Canaries and large-graph volume

Cora canary checks: HARP suppresses planted/top-decile confidence cues relative to none/LBP (top-decile conf AUROC 0.551/0.548/0.266 for none/LBP/HARP; planted 0.527/0.543/0.423); population LiRA remains primary. Table XIV: on ogbn-arxiv ($\sim 169k$; NeighborLoader; gate off, warmup 3) [28], strong LBP drops Acc to 0.179 [0.175, 0.184]. HARP recovers 0.599 [0.597, 0.601] (paired $\Delta\text{Acc} +0.420$ [0.414, 0.425]; 3 seeds) at $\text{Frac}=0.40$. Undefended LiRA stays near chance at $n_{sh} \in \{2, 4\}$ and also at $n_{sh}=16$ (LiRA 0.510; Acc 0.666)—so the Acc recovery is a serving result under an audit-null regime, not under-powered shadows. On ogbn-products ($\sim 2.4M$ nodes) [28], Table XIII reports a serving probe plus BFS LiRA at $n \in \{15k, 40k\}$ and $n_{sh} \in \{2, 4, 8\}$. Full-graph products LiRA OOMs on this host; we do *not* claim a million-node privacy audit. Across standard-recipe probes LiRA stays near chance while HARP recovers Acc vs. strong LBP (e.g., $15k/n_{sh}=8$: HARP Acc 0.693 vs. LBP 0.343).

a) *Canary-stressed probe (above-chance signal at scale)*.: Standard products/arxiv recipes sit near chance, so Acc recovery alone is a weak privacy claim. We plant 300 unique-feature canaries on the 15k BFS probe (half in train / half held out; $n_{sh}=4$; 3 seeds): undefended canary-LiRA is ≈ 1.00 while population LiRA is 0.593 (Table XIII). HARP cuts canary-LiRA to 0.891 (still high) at Acc 0.685 with $\text{ExactFrac}=0.60$; strong LBP reaches 0.805 only by collapsing Acc to 0.379 at $\text{ExactFrac}=0$. Population LiRA barely moves (≈ 0.59) under either release defense—defense-aware shadows recalibrate—so the scale takeaway is Acc-preserving ExactFrac under a hard canary stress, not a population-LiRA win at 15k. We do *not* claim a full 2.4M-node privacy audit.

If audit LiRA ≈ 0.5 , prefer $\text{Frac}=0$; if LiRA $\gg 0.5$ or $\text{ExactFrac} \geq c$ is required, set Frac on val Acc.

b) *Operator workflow*.: Audit with defense-aware LiRA ($n_{sh} \geq 16$ when affordable); if LiRA ≈ 0.5 set $\text{Frac}=0$; else CFS on val Acc under (c, τ) (Table IV); prefer ensemble constructors + optional DCS for slice risk; enforce shared $B=1$ under Sybil velocity; use HARPoGAP when a legal ε is required.

TABLE XIII

OGBN-PRODUCTS: STANDARD BFS LADDER VS. CANARY-STRESSED 15K
($n_{sh}=4$, 3 SEEDS). CAN. = CANARY LiRA.

Setting	Acc	LiRA	Can.	EF
Full load none (2.45M)	0.695	—	—	1.00
Std. 15k none/LBP/HARP	0.876/0.343/0.693	0.528/0.501/0.513	—	1/0/0.60
Canary none	0.861	0.593	0.997	1.00
Canary LBP $b=0.3$	0.379	0.581	0.805	0.00
Canary HARP	0.685	0.592	0.891	0.60

TABLE XIV

OGBN-ARXIV VOLUME (3 SEEDS). LiRA_{2/4} ARE DEFENSE-AWARE;
UNDEFENDED LiRA AT $n_{sh}=16$ IS 0.510 (NEAR-CHANCE CREDIBILITY
CHECK).

Def.	Acc	LiRA ₂	LiRA ₄	Mass
none	0.666	0.502	0.503	—
LBP	0.179	0.498	0.501	50 803
HARP	0.599	0.514	0.505	20 322

VII. DISCUSSION

HARP is a Big-Data score-serving mechanism for $\text{ExactFrac} \geq c > 0$ —not a claim to beat uniform noise when $c=0$, and not an equation of ReplayFrac with raw-posterior equality. The venue fit is prediction-API infrastructure under Volume/Velocity/Veracity: selective release, cache/replay accounting, and audits on high-volume graphs—membership privacy enters as an operator dial on that serving stack. ExactFrac formalizes a raw-score SLA that generic response caches do not automatically provide [40], [41]; $c=0.60$ is a study default, and sticky/global/seeded noise are not substitutes (Table VIII). GAP-agg already meets $\text{ExactFrac}=1$ with finite (often large) ε ; HARP’s complementary claim is high-Acc ExactFrac without training DP, plus constructors/CFS/slice controls (Prop. 2). **Limits:** (i) $\text{ExactFrac} > 0 \Rightarrow$ infinite global ε on released scores; (ii) clean-slice AUROC stays above chance at $c=0.60$ —structural; (iii) CFS τ is an operator dial that must track the LiRA budget; (iv) products privacy is a BFS/canary ladder; (v) GAP-agg/MemGuard are reimplementations; (vi) serving benches are in-process microbenchmarks, not production traces. Future: learned constructors, distributed full-graph LiRA, production API traces.

VIII. CONCLUSION

HARP equips GNN prediction APIs with ExactFrac -constrained score serving: selective release meets raw-posterior $c > 0$ where fresh and stateful-noisy uniform policies cannot, constructors cut clean-slice attack AUROC, CFS configures Frac under operator (c, τ) , and HARPoGAP preserves training DP for public-metadata constructors. Headline evidence uses $n_{sh}=16$; residual clean-slice risk and scoped large-graph Acc/serving probes are reported honestly. Use training-time DP (alone or under HARP) when a legal global ε is required.

a) *Reproducibility*.: Code (defenses/harp.py, run_nsh16_headline.py, run_cfs_nsh16.py, run_exactfrac_sla_evidence.py,

run_products_canary_leakage.py, run_bulletproof.py), locked configs, and frozen CSVs: <https://github.com/krishharma/privacy-gnn>. Headline LiRA $n_{\text{shadows}}=16$; CFS uses $\tau=0.70$ on validation; RawExactFrac SLA evidence in Table VIII.

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